Production Order Create/Edit — Screen Processing Design (画面処理設計)

> **Revision of 2026-10-24 (WI-017).** This file supersedes `001_DD-SPD_製造指示登録・編集.md`, which stays unchanged as the earlier approved revision. Change: the production-order notes limit is 1000 characters (REQ-090, supersedes the 500 of REQ-016); see `work-items/WI-017/`.

001_DD-SPD — elaborates 001_DD, implements 001_BD (SCR-001), requirements REQ-010–REQ-019.

Document control (改版履歴)

Field	Value
Document ID	001_DD-SPD
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-002
Created by	Claude (for ThanhTN)
Created date	2026-09-18
Last updated by	Claude (for ThongTM)
Last updated date	2026-10-24

Version	Date	Author	Revision content
1	2026-09-18	Claude (for ThanhTN)	Initial creation. Moves 001_DD revision 1's processing flows P-01–P-05 into this document and splits them per component
2	2026-09-22	Claude (for ThanhTN)	WI-004 (DEC-016): the page renders inside the shared `AppHeader`, which now contains the navbar (`AppNavbar`, designed in 003_DD module 10 and 003_DD-SPD §8; breadcrumb links become `GuardedLink`, 003_DD module 11). No processing step of this screen changes. WI-004 DEC-022: the form registers its dirty state with the shared `NavigationGuard` (003_DD module 11, 003_DD-SPD §9), so navbar, breadcrumb and app-name links open the existing discard dialog with a pending destination (001_BD v7 E-07a); Discard navigates there
3	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. Page titles, button names and the dialog title quote the Japanese text with an English gloss (WI-005 DEC-003). No field, rule, API or processing changes

Version	Date	Author	Revision content
4	2026-10-24	Claude (for ThongTM)	WI-017 (REQ-090–REQ-092): notes limit 500 → 1000 characters. Notes counter `{n}/1000`, red above 1000. No other processing change. Dated file name per WI-017 DEC-002

Overview and process list

Field	Value
Screen / file name	`src/frontend/src/features/production-orders/ProductionOrderPage.tsx` and the components it composes; the server side of each request is in `ProductionOrdersController`
Overview	How SCR-001 loads, validates, saves, cancels and reacts to every API outcome, one block per component

No	Process name	Overview	Notes
1	`ProductionOrderPage` — P-01 Load	Role gate, parallel fetch, page state, title, flash message	E-01
2	`ProductionOrderForm` — P-02 Create / P-03 Update (client side)	Blur/submit validation, request build, response handling	E-02–E-06, E-08
3	`ProductionOrderForm` — P-05 Save-response handling	Mapping of each HTTP outcome to UI	E-06
4	`DiscardChangesDialog` — P-04 Cancel	Dirty check and confirmation	E-07, E-09
5	`ProductionOrdersController` / `ProductsController` — request boundary	Auth, content type, dispatch to the service, result → HTTP	Server steps are in 001_DD-FN

Reference documents

No	Document	Purpose / use	Notes
1	001_DD	Items, states, message catalog, module list	Parent document
2	001_DD-API	Endpoint contracts called below	
3	001_DD-FN	Server-side steps (`CreateAsync` , `UpdateAsync` , …)	
4	001_BD	Events E-01–E-09, validation V-01–V-08	

Processing design

1. ProductionOrderPage — P-01 Load

Field	Value
Detail	Route component for `/production-orders/new` and `/production-orders/:id`
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: decide the mode from the route, apply the client-side role gate, fetch the data, then either render the form or a full-page panel. Runs on mount and on **再読み込み** (Reload) / **Try again**.

Used components / services

No	Name	Overview	Notes
1	`useAuth`	Current user and roles	WI-001
2	`api.ts` (`listProducts` , `getOrder`)	Typed wrappers over `apiClient`	001_DD module 10
3	`ProductionOrderForm` , `MessageBanner`	Rendered when ready	
4	`useLocation` / `useNavigate`	Flash message from the create redirect	react-router

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Mode = `create` if the route is `/new` , else `edit` with `id`	—	—	—
2	Role gate on `user.roles`	Neither `Admin` nor `Operator` → stop	—	`forbidden` panel (MSG-E012), no API call
3	Set state `loading` , `document.title` to the mode's title (the order number is added after load)	—	—	Skeleton card, `aria-busy="true"`
4	Fetch in parallel: products, and in edit mode the order	—	`GET /api/products` ; `GET /api/production-orders/{id}`	—

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
5	Order response	404 → `notFound`; 401 → apiClient redirects to `/login`; 403 → `forbidden`; network/5xx → `loadError`	apiClient	Not-found (MSG-E011) / forbidden / load-error (MSG-E013 + **Try again**) panel, focus moved to the panel
6	Products response	Failure → `loadError`; empty list → continue with no options	apiClient	Product select shows MSG-E014 and is disabled
7	Build `initialValues`: from the order (edit) or empty with status `Draft` (create); set title `製造指示 {orderNumber} – ProductionManagementAI` (Production order {orderNumber}) in edit mode; create mode's title is `新規製造指示 – ProductionManagementAI`	—	—	`ready`
8	If `location.state.flash` is set (after create), show that message and clear the state with `navigate(location.pathname, { replace: true, state: null })`	No flash → skip	—	Success banner MSG-I001

2. ProductionOrderForm — P-02 Create / P-03 Update (client side)

Field	Value
Detail	Controlled form for items 6–16, owning values, errors, dirty state and the save request
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: validate on blur for early feedback, validate everything on Save, then send one request. The server repeats every check (001_DD-FN), so the client checks are advisory. After a successful update the form re-bases on the server response, so `isDirty` becomes false and the lock/status options reflect the new status.

Used components / services

No	Name	Overview	Notes
1	`validation.ts`	`validateProduct`, `validateQuantity`, `validateDueDate(value, initial, mode)`, `validateNotes` → message ID or null	Mirrors V-01–V-05
2	`messages.ts`	Message ID → text	001_DD catalog
3	`api.ts` (`createOrder`, `updateOrder`)	POST/PUT wrappers	001_DD-API §2, §4
4	`FieldError`, `MessageBanner`	Error display	

Processing flow — blur (E-02) and status change (E-08)

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Field loses focus	Read-only field → skip	—	—
2	Run that field's validator	Due date: check "≥ browser-local today" only in create mode or when the value differs from `initialValues.dueDate` (DEC-009, DEC-021)	`validation.ts`	Inline error with `aria-invalid`, or the error cleared
3	Notes input (E-03)	—	—	Counter `{n}/1000` (code points); red above 1000
4	Status change	Options are only `[current, ...allowedNextStatuses]`	—	Value changes locally; nothing saved until Save

Processing flow — Save (E-04)

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Run all validators (read-only fields skipped)	Any error → show all, focus the first invalid field in DOM order, stop	`validation.ts`	Validation-error state
2	Build the body: `productId`, `quantity` (as number), `dueDate`, `notes` (trimmed, `""` → `null`); in edit mode also `status` and `version` from `initialValues`	—	—	—
3	Set `saving`: Save disabled, label 「保存中…」 (Saving…)	—	—	Saving state
4	Send	Create → POST; Edit → PUT	`createOrder` / `updateOrder`	—
5	Create success (201)	—	`navigate('/production-orders/{id}', { replace: true, state: { flash: 'MSG-I001' } })`	Page re-runs P-01 in edit mode; `replace` keeps Back from returning to an empty create form
6	Update success (200)	—	—	`initialValues = values = response`; banner MSG-I002 (`role="status"`); lock and status options recomputed from `isProductQuantityEditable` / `allowedNextStatuses`
7	Failure	—	Process 3 (P-05)	—

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
8	Clear `saving`	always	—	Save enabled again

3. ProductionOrderForm — P-05 Save-response handling (E-06)

Field	Value
Detail	Maps every non-success save outcome to UI
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: an `ApiError` from apiClient carries the status and the parsed Problem Details. Entered values are always kept, except after a 401 (session gone) or when the user presses **Reload** after a 409.

Used components / services

No	Name	Overview	Notes
1	`apiClient` `ApiError`	`{ status, problem? }`	001_DD module 10
2	`messages.ts`	IDs → text	

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	400 `VALIDATION`	For each key in `problem.errors` → that field's message	—	Inline errors, focus first invalid
2	401	—	apiClient `onUnauthorized`	Auth cleared → `ProtectedRoute` → `/login` (unsaved values lost; accepted)
3	403	—	—	Forbidden panel (MSG-E012)
4	404 `MSG-E011`	—	—	Not-found panel
5	409 `MSG-E009`	—	—	Error banner with **再読み込み** (Reload) → P-01 (discards local edits)
6	422 `MSG-E007` / `MSG-E008`	—	—	Error banner with that message

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
7	415, 5xx, network error	—	—	Error banner MSG-E013; Save enabled again

4. DiscardChangesDialog — P-04 Cancel (E-07, E-09)

Field	Value
Detail	Confirmation before discarding unsaved changes (REQ-019, DEC-018)
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: Cancel checks `isDirty` (values compared with `initialValues`, notes trimmed). Only a dirty form opens the dialog. Covers the Cancel button only, not browser Back or tab close (DEC-018).

Used components / services

No	Name	Overview	Notes
1	Native `<dialog>` + `showModal()`	Focus trap, Escape, `::backdrop`	No library
2	`useNavigate`	Leave to `/production-orders`	SCR-002, since WI-003

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Cancel clicked	`isDirty` false → `navigate('/')`	—	Home
2	Open the dialog	`isDirty` true	`dialog.showModal()`	Dialog 「変更を破棄しますか？」 (Discard your changes?); focus on **編集を続ける** (Keep editing)
3	**破棄** (Discard)	—	`dialog.close()`, `navigate('/')`	Home; nothing saved
4	**編集を続ける** (Keep editing) or Escape (`cancel` event)	—	`dialog.close()`	Form with all values; focus returns to 「キャンセル」 (Cancel)

5. ProductionOrdersController / ProductsController — request boundary

Field	Value
Detail	HTTP entry points for 001_DD-API; no business logic
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: the ASP.NET Core pipeline authenticates and authorizes, then checks the content type. The action calls one service method and maps its `Result` to HTTP. All business checks happen in 001_DD-FN.

Used components / services

No	Name	Overview	Notes
1	Policy `ProductionOrderEditor`	`RequireRole("Admin", "Operator")`	DEC-001
2	`[Consumes("application/json")]`	415 on other content types	DEC-020
3	`ProductionOrderService`	Use cases	001_DD-FN
4	`ProblemResults` helper	`Result` → Problem Details (`type` , `title` , `status` , `code` , `errors` , `traceId`)	DEC-023

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Authentication (cookie)	None → 401, empty body	ASP.NET Core	—
2	Authorization (policy)	Missing role → 403, empty body	ASP.NET Core	—
3	Route match `{id:guid}`	Not a GUID → 404	routing	—
4	Content type (POST/PUT)	Not JSON → 415	`[Consumes]`	—
5	Model binding	Binding error → 400 with message IDs (custom `InvalidModelStateResponseFactory`)	MVC	—

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
6	Call the service method	—	`ListProductsAsync` / `GetAsync` / `CreateAsync` / `UpdateAsync`	—
7	Map the result	`Ok` → 200 (201 + `Location` for create); `Invalid` → 400; `NotFound` → 404; `Conflict` → 409; `RuleViolation` → 422	`ProblemResults`	Response
8	Unhandled exception	—	`AddProblemDetails` + exception handler	500 MSG-E013, no exception detail

Unresolved decisions

None. Related decisions: DEC-007, DEC-008, DEC-009, DEC-018, DEC-020, DEC-021, DEC-023 in `work-items/WI-002/decisions.md`.